TrustMyBike

IoT Algorithm and Services

Tiziano Coluzzi | Aldo Bosco | Alessandro Arellano Altuna

WHAT WE DID

Data Collection

Raw data collection, mean filter application, and angle calculation to normalize gravity acceleration.

Machine Learning

Training phase and edge inference implementation to detect anomalies (Outlier Detection for potholes).

IoT Algorithm

Code engineering for the efficient management of dynamic thresholds and static baseline values.

Hardware Setup

Design and testing of the first continuous power supply stage based on the bicycle's dynamo.

ALGORITHM FOCUS

Sampling Frequency

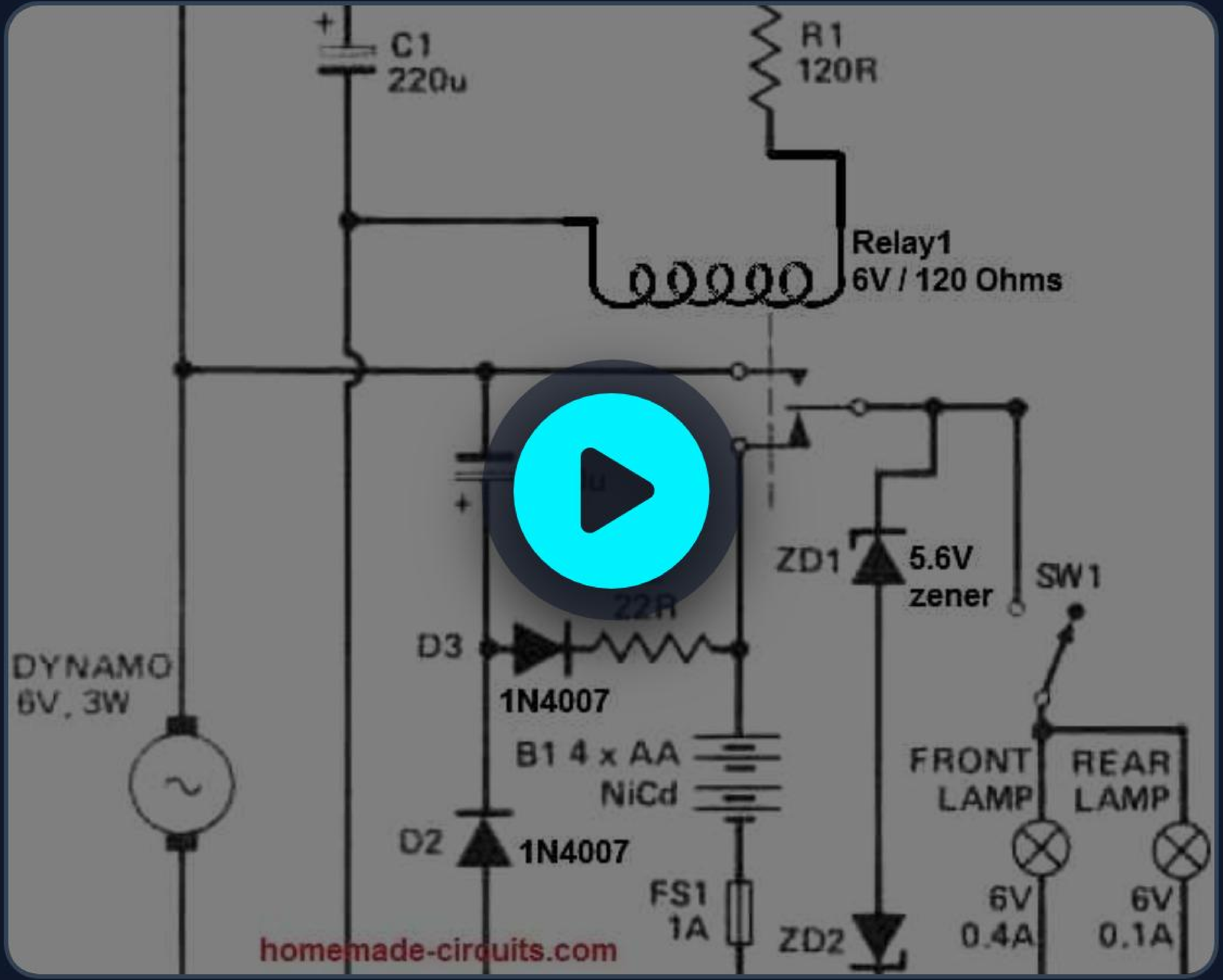
Initially calculated by running the **FFT** on the public "bike&safe" dataset. The goal is to gather data from our own system and dynamically adjust the frequency according to velocity.

Pothole Detection

Intensive use of **Machine Learning** algorithms for Outlier Detection. The model is optimized to analyze normalized data, isolating specific impact peaks.

DYNAMO SETUP: 1ST TEST

- Practical road test to verify the actual output of the 6V/3W dynamo.
- Difficulties encountered in managing variable Alternating Current (AC).
- Recording of anomalous voltage peaks preventing stable ESP32 power supply.
- Visual analysis of the failure in the first rectification and condensation stage.



PROBLEMS & SOLUTIONS

STATUS	DETECTED PROBLEM (ROAD TEST)	ADOPTED / PLANNED SOLUTION
✓	The same pothole was detected multiple times.	Implementation of Multi-Threshold logic (separate entry/exit thresholds).
✓	Continuous downhill mistaken for a "mega-pothole".	Gravity removal and transition to normalized accelerometer data.
!	Huge drift in speed and angle calculation.	(Open issue) MPU6050 integration drift. New hardware sensor required.

FUTURE WORK

Road Quality

Creation of a continuous predictive model to evaluate road surface quality.

Input Optimization

Automatic FFT calculation and max acceleration in pre-processing.

Hall Sensor

New Hardware: Definitively solves the speed calculation drift problem.

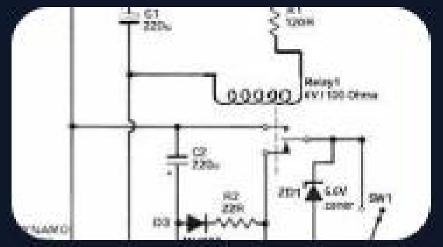
Crowdsourcing

Massive GPS data integration via app for shared community mapping.

Power Monitor

Installation of an INA sensor for live battery telemetry.

IMAGE SOURCES



https://www.homemade-circuits.com/wp-content/uploads/2023/09/bicycle-dynamo-charger-circuit-with-relay-changeover.jpg

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